AIIDE 2026 · 22nd AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment

Generating Board Games with MCTS and LLM Agents

Gabriel Bauer de Oliveira · Anderson Rocha Tavares

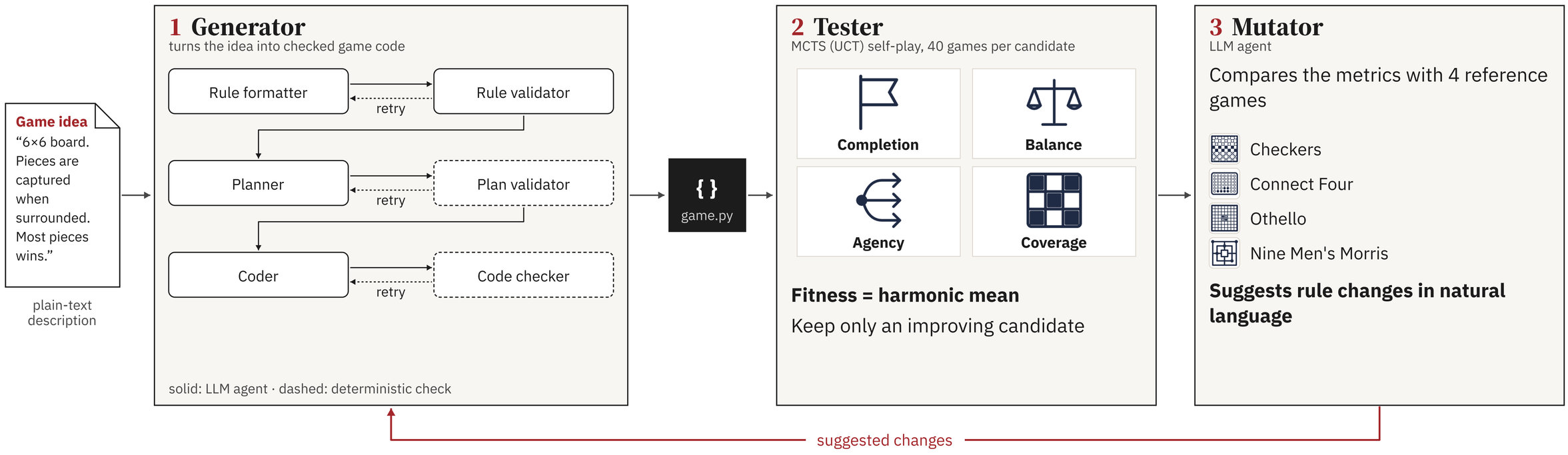
Instituto de Informática, Universidade Federal do Rio Grande do Sul (UFRGS), Porto Alegre, Brazil · {gbboliveira, artavares}@inf.ufrgs.br

From a plain-text game idea to a playable, tested board game

LLM agents write and revise the game as Python code, MCTS agents play it, and the results of play decide what to change next.

Input a game idea in plain text → **Loop** generate → test → mutate → **Output** the best playable game found, in Python

Approach: generate, test, mutate, repeat



Experimental setup

Games

- **C1** Connection game with bridge tiles
- **C2** Capture by surrounding, 6×6 board
- **C3** Asymmetric pursuit: King vs 4 Guards
- **C4** Promotion game + a mechanic the LLM invents

Models

- DeepSeek-V4-Pro
- Claude-Sonnet-5
- GPT-5
- Qwen3.6-Plus

Protocol

1. One run per model–game pair: 4 LLMs × 4 games = 16 runs.
2. Each candidate is played 40 times by two MCTS agents (1 s per move, 60-turn cap).
3. A candidate is kept only if it closes 4% of the gap to 1.0, gains 50%, or makes balance defined, and does not lose on both completion and balance.
4. Run stop: after 2 rejections in a row or 10 10 cycles.

Metrics of reference games

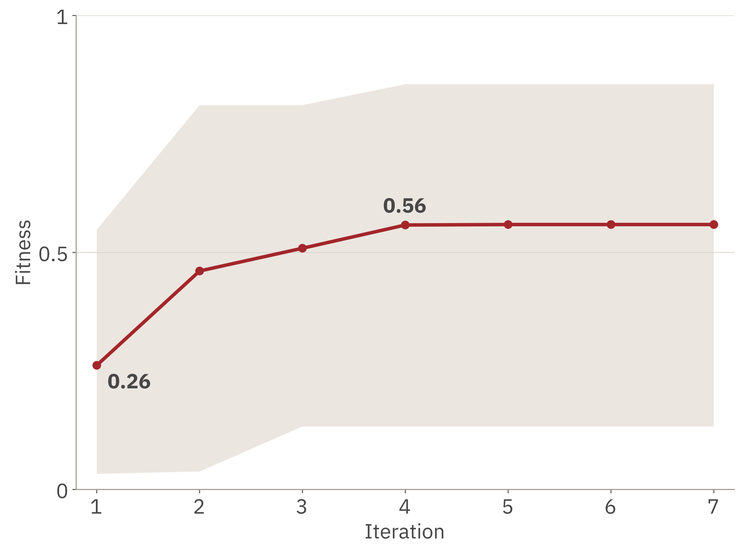
	Compl.	Balance	Agency	Cover.	Fitness
Checkers	0.363	0.997	0.824	0.499	0.573
Connect Four	0.954	0.977	0.981	0.741	0.901
Othello	0.968	0.994	0.966	0.998	0.981
Nine Men's Morris	0.699	0.933	0.998	0.986	0.885

Results

Final fitness, game × model

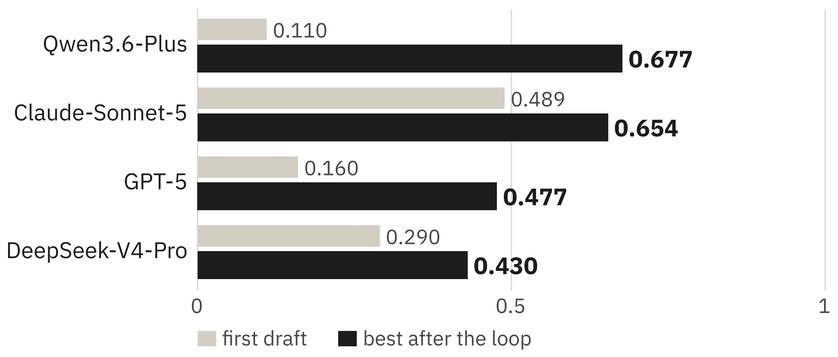
	DeepSeek V4-Pro	Claude Sonnet-5	GPT-5	Qwen3.6 Plus	Mean
C1 Connection	0.819	0.784	0.299	0.817	0.680
C2 Capture	0.010	0.858	0.700	0.982	0.637
C3 Pursuit	0.038	0.165	0.038	0.038	0.070
C4 Promotion	0.854	0.809	0.868	0.870	0.850
Mean	0.430	0.654	0.477	0.677	

Fitness of the best game so far, all runs

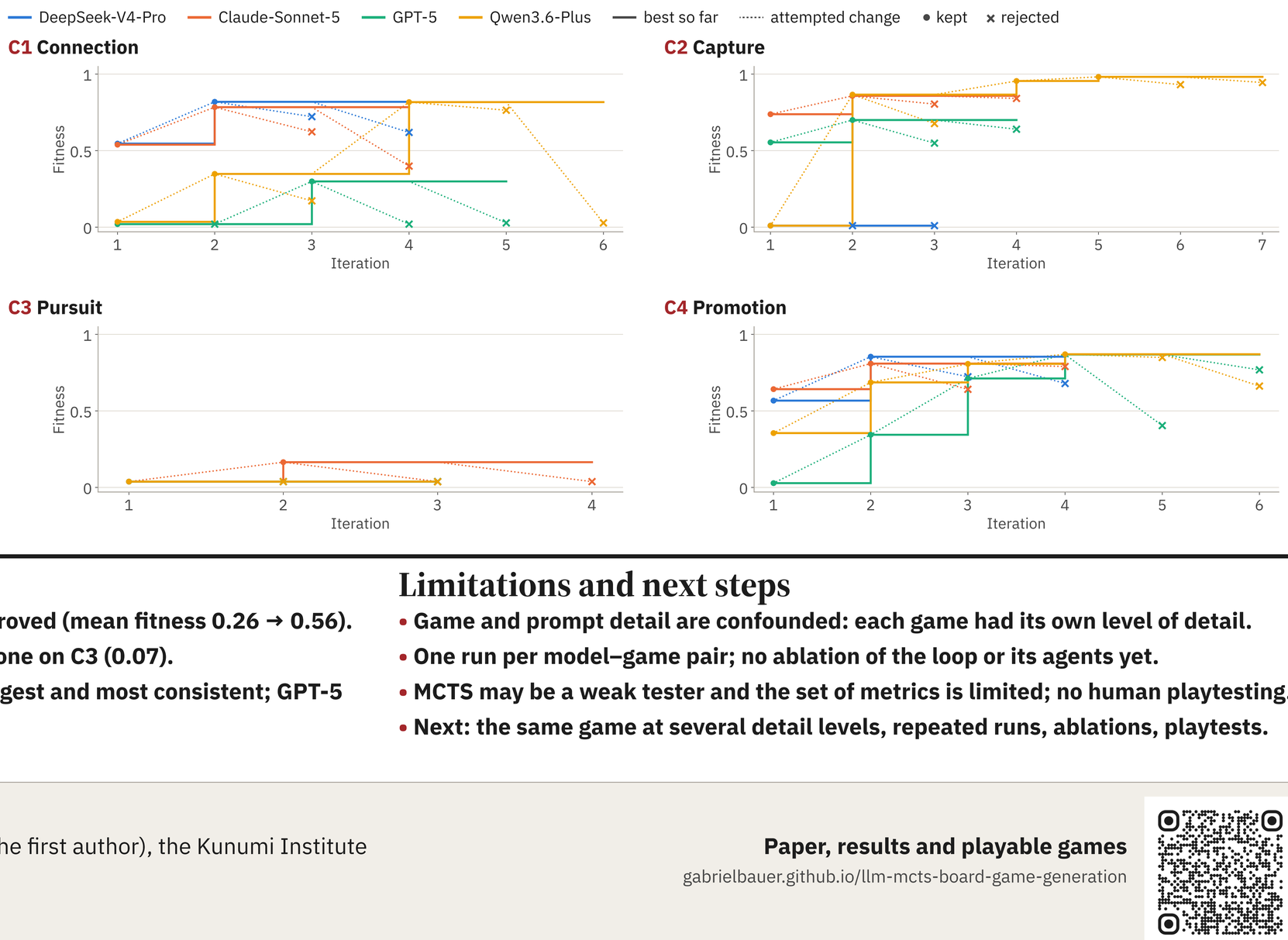


Line: mean of the 16 runs. Band: middle half of the runs (25th–75th percentile).

First draft vs best game



Fitness of the best game so far, per game



Conclusions

- **The loop helps the improvement: 12 of 16 runs improved (mean fitness 0.26 → 0.56).**
- **The game matters: balanced games on C4 (0.85), none on C3 (0.07).**
- **Claude-Sonnet-5 and Qwen3.6-Plus were the strongest and most consistent; GPT-5** and DeepSeek-V4-Pro varied more.

Limitations and next steps

- **Game and prompt detail are confounded: each game had its own level of detail.**
- **One run per model–game pair; no ablation of the loop or its agents yet.**
- **MCTS may be a weak tester and the set of metrics is limited; no human playtesting.**
- **Next: the same game at several detail levels, repeated runs, ablations, playtests.**

Acknowledgments

This work was supported by FAPERGS (scholarship to the first author), the Kunumi Institute and CAPES (Finance Code 001).

FAPERGS logo · Kunumi logo · CAPES logo

Paper, results and playable games

gabrielbauer.github.io/llm-mcts-board-game-generation